

Education		
Jan 2026 – current	Master of Mathematics, Thesis, Computer Science , University of Waterloo, ON, Canada. Coursework: Adv. Topics in SE: Large Language Models for SE, Adv. Topics in SE: SE for Big Data and AI. Supervisor: Dr. Raouf Boutaba, Grade: 99.5/100	
Sep 2021 – May 2025	Bachelor of Science, Computer Science , Lahore University of Management Sciences (LUMS), Pakistan. Graduated with High Distinction. GPA: 3.97/4.0	
Jun 2023 – Aug 2023	Summer Session , Stanford University, CA, USA. Coursework: CS 229 - Machine Learning, ENGR 145 - Technology Entrepreneurship. GPA: 4.0/4.3	

Professional Experience		
Jan 2026 – current	Graduate Research Assistant , <i>University of Waterloo</i> . As part of a cross-functional team of CS and ECE departments, I abstracted a testbed design for terrestrial-non-terrestrial 5G integration that utilizes cross-layer information sharing to drive optimizations across scheduling, beamforming, and transport. This testbed diagram was utilized to explain our solution during stakeholders meetings and I led weekly project presentations for the CS team. Utilizing my experience with agentic coding, I mapped the testbed proposals onto open-source 5G stacks which enabled the ECE team to implement non-terrestrial channel models. Skills: 5G Core, O-RAN, Open5GS, srsRAN (OCUDU), OpenAirInterface, GNU Radio, channel emulation.	
Aug 2025 – Jan 2026	Software Engineer , <i>GreatFrontEnd</i> , Singapore, Remote, Full-time. I led the development of a novel layout generation pipeline with Gemini’s multimodal models and OpenCV components by drawing ideas from published research on automated layout generation. I also coded and deployed a scalable, real-time artifact search engine on Railway utilizing vector and lexical search , scaling to > 110k data points with search latencies under 50ms . Finally, I supervised a SWE intern on technical content writing for front-end engineering audiences on the company’s LinkedIn. Skills: LLM pipelines and orchestration, product design, semantic search, vector databases, OCR, OpenCV.	
Feb 2025 – Apr 2025	Software Engineering Intern , <i>GreatFrontEnd</i> , Singapore, Remote, Part-time. I implemented advanced DOM manipulation using event listeners to enable an executable JS IDE on the web which boosted user interaction on the company’s ed-tech website. I also curated over 200 SWE interview questions in JavaScript and authored effective technical content for front-end engineering audiences, driving unprecedented growth in LinkedIn engagement . Skills: React, TypeScript, Next.js, tRPC, Prisma, Framer Motion, technical content writing.	
May 2024 – Aug 2024	Software Engineering Intern , <i>AI Centre for Educational Technologies</i> , Singapore, On-site, Full-time. Engineered SoftMark, a productionized digital grading platform deployed at NUS and secondary schools across Singapore. I implemented a trie-based algorithm over open-source OCR to detect question numbering on answer sheets, doubling teachers’ productivity by filtering unattempted questions. I also created an intuitive 2D grid allocation component to replace a linear menu, enabling fairness in grading workload distribution among graders. Skills: React, TypeScript, Golang, PostgreSQL, Material UI	
Jul 2022 – Aug 2022	Exchange Semester , <i>University of Baltistan</i> , Skardu, Pakistan. Coursework: SCI 204 - Water Resources Management. I surveyed and presented solutions to improve water supply distribution and fairness in the water-scarce Skardu valley of northern Pakistan through field interviews and Google Earth satellite imagery. I presented an actionable report to local stakeholders .	

Research		
Sep 2024 – Apr 2025	Undergraduate Senior Year Project , <i>TweakLLM: A Routing Architecture for Dynamic Tailoring of Cached Responses</i> . Programmed and benchmarked an adaptive caching architecture for LLM systems that utilizes a lightweight model to dynamically adapt cached responses to serve similar user queries, reducing response generation latency and cutting compute costs by up to 68% on real-world user-LLM interaction datasets. The system utilizes semantic caching by integrating Milvus as a vector database. Link to paper: arXiv:2507.23674 . Advised by Dr. Zafar Ayyub Qazi, Dr. Ihsan Ayyub Qazi.	
Feb 2024 – May 2024	Course Research Project , <i>Optimizing Vector Databases for Audio-Based RAG LLMs</i> , CS678 - Topics in Internet Research, LUMS. Benchmarked Milvus, Weaviate, and Qdrant on audio data using various embedding schemes, similarity metrics, and indexing methods to determine the optimal configuration for real-time audio-based interaction in LLMs .	
Sep 2023 – May 2024	Directed Research , <i>Improving Federated Learning for Causal Graph Discovery</i> . Analyzed the trade-off between accuracy and privacy within state-of-the-art algorithms (FedPC, FedFC, etc.) for federated causal graph discovery . Mentored by Dr. Ihsan Ayyub Qazi at LUMS.	
Jan 2024 – Mar 2024	Directed Research , <i>Tuneable Consistency Edge Datastore to Support Stateful Mobile Applications over 5G</i> . My team studied a state migration algorithm to minimize application down-time during base station handovers for mobile edge applications. As a junior researcher in the team, I assimilated literature review and whiteboarded how to integrate a fault-tolerant edge data-store with tunable consistency for state migration. Mentored by Dr. Zafar Ayyub Qazi at LUMS.	

Achievements and Honors		
Jun 2026	Third Place in University of Waterloo’s Future Cities Innovation Challenge	<i>University of Waterloo</i>
2025 – 2027	International Master’s Award of Excellence valued at C\$12,500	<i>University of Waterloo</i>
2022 – 2025	Placed on Dean’s Honor List	<i>Lahore University of Management Sciences</i>
2022 – 2025	50% Tuition Merit Scholarship	<i>Lahore University of Management Sciences</i>
2024 – 2025	Varsity Swim Team	<i>Lahore University of Management Sciences</i>
2022 – 2023	Additional Director Marketing	<i>The Music Society of LUMS</i>
2023	70% Tuition Waiver Award	<i>Stanford Summer Session</i>
2019 – 2021	100% Merit Scholarship Award in A-Level	

Feb 2020

Champions, Lahore Regional - National Science Bowl Pakistan

2020

Host, TEDxYouth@BeaconhouseLahore

Technical Skills

Web-dev

React, Next.js, Redux, Recoil, Material UI, tRPC, Prisma ORM

Other tools

Git, Docker, AWS, Railway, Meilisearch, Milvus, Supabase, Open5GS, srsRAN, GNU Radio, ns-3